

In [1]:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from sympy import *
from scipy import stats
from datetime import timedelta
import matplotlib.ticker as ticker
from sklearn import metrics
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn import datasets
from sklearn import svm
from sklearn.model_selection import cross_validate
import nltk
from nltk.tokenize import sent_tokenize
from sklearn.model_selection import cross_val_score
from scipy.stats import ttest_ind
import seaborn as sns

```

In [2]:

```
data = 'D:/AI Course/AI Capstone Project/UpdatedTemporaryStaffList.csv'
```

In [3]:

```
Data = pd.read_csv(data)
```

In [4]:

Data

Out[4]:

	RowID	AssignmentID	OrderID	Title	StartDate	ProjectedEndDate	ActualEndDate	AssignmentStatus
0	29605	1001	1001	NaN	26/12/2011	31/08/2013	23/01/2015	E
1	29606	1002	1002	NaN	05/09/2012	31/08/2013	13/09/2013	E
2	29607	1003	1003	NaN	02/04/2007	31/08/2013	30/09/2015	E
3	29608	1004	1004	NaN	23/09/2009	31/08/2013	23/01/2014	E
4	29609	1005	1005	NaN	23/09/2009	31/08/2013	23/01/2014	E

In [5]:

```
Data.isnull().sum()
```

Out[5]:

RowID	0
AssignmentID	0
OrderID	0
Title	6662
StartDate	0
ProjectedEndDate	0
ActualEndDate	0
AssignmentStatus	0
StandardRate	0
ChargeRate	0
HoursPerWeek	0
HoursInvoicedToDate	0
JobTitleCode	11
SupplierName	0
CreatedDate	0
CreatedBy	0
UpdatedDate	0
UpdatedBy	0
dtype:	int64

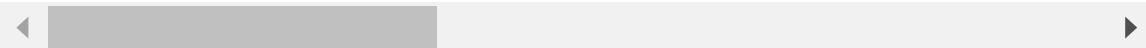
In [6]:

```
Data.drop(['Title','JobTitleCode'], axis=1)
```

Out[6]:

	RowID	AssignmentID	OrderID	StartDate	ProjectedEndDate	ActualEndDate	Assig
0	29605	1001	1001	26/12/2011	31/08/2013	23/01/2015	
1	29606	1002	1002	05/09/2012	31/08/2013	13/09/2013	
2	29607	1003	1003	02/04/2007	31/08/2013	30/09/2015	
3	29608	1004	1004	23/09/2009	31/08/2013	23/01/2014	
4	29609	1005	1005	23/09/2009	31/08/2013	23/01/2014	
...	...	...	...	...	...	...	
6657	37732	SA562107	RQ858526	17/04/2023	10/07/2023	10/07/2023	
6658	37733	SA563163	RQ859615	25/04/2023	10/07/2023	20/07/2023	
6659	37734	SA559608	RQ859261	26/04/2023	10/10/2023	26/10/2023	
6660	37735	SA564031	RQ766531	09/05/2023	21/04/2023	31/08/2023	
6661	37736	SA565938	RQ899529	03/05/2023	03/07/2023	03/07/2023	

6662 rows × 16 columns



In [7]:

```
#create feature variable
feature_data = Data[["StandardRate","ChargeRate","HoursPerWeek","HoursInvoicedToDate"]]

#"StartDate","ProjectedEndDate","ActualEndDate"
```

In [8]:

```
#select target variable what am i trying to workout
target_data = Data[["AssignmentStatus"]]
```

In [9]:

```
x = feature_data
y = target_data
```

In [10]:

```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2,random_state=4)

print(x_train.shape)
print(x_test.shape)
```

```
(5329, 4)
(1333, 4)
```

In [11]:

```
logreg = LogisticRegression()
logreg.fit(x_train,np.ravel(y_train))
```

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\linear_model_logistic.py:762: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html> (<https://scikit-learn.org/stable/modules/preprocessing.html>)

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression (https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression)

```
n_iter_i = _check_optimize_result(
```

Out[11]:

```
LogisticRegression()
```

In [12]:

```
#create the model to predict results
y_pred = logreg.predict(x_test)
```

In [13]:

```
#test train split on the logistic regression model
metrics.accuracy_score(y_test, y_pred)
```

Out[13]:

```
0.8439609902475619
```

In [14]:

```
#cross validation result for Logistic regression
cv1 = cross_validate(logreg, x, y, cv=5)
print(cv1['test_score'].mean())
```

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection_split.py:670: UserWarning: The least populated class in y has only 1 members, which is less than n_splits=5.

warnings.warn("The least populated class in y has only %d"

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\utils\validation.py:72: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

return f(**kwargs)

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\linear_model_logistic.py:762: ConvergenceWarning: lbfgs failed to converge (status=1): STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html> (<https://scikit-learn.org/stable/modules/preprocessing.html>)
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic

In [15]:

```
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=27)
knn.fit(x_train, y_train)
```

<ipython-input-15-8865816bf805>:3: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

knn.fit(x_train, y_train)

Out[15]:

KNeighborsClassifier(n_neighbors=27)

In [16]:

```
#test train method for knn
knn.score(x_test, y_test)
```

Out[16]:

0.900225056264066

In [17]:

```
#create a model to okr out the best n for knn
k_range = range(1,40)

scores = []

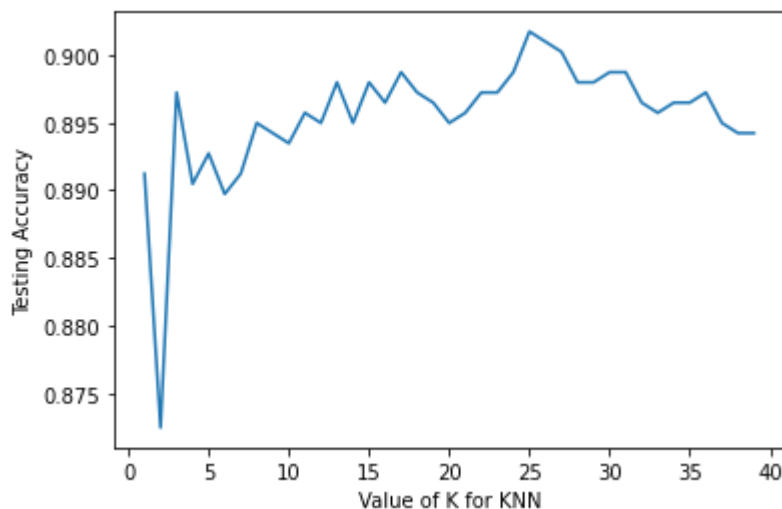
for k in k_range:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(x_train, np.ravel(y_train))
    y_pred = knn.predict(x_test)
    accuracy = metrics.accuracy_score(y_test, y_pred)
    scores.append(accuracy)
```

In [18]:

```
#plot out the possible knn neighbors and see which would perform best
plt.plot(k_range, scores)
plt.xlabel("Value of K for KNN")
plt.ylabel("Testing Accuracy")
```

Out[18]:

Text(0, 0.5, 'Testing Accuracy')



In [19]:

```

#create new knn model
knn_cv = KNeighborsClassifier(n_neighbors=27)

#train new model with folds
cv_scores = cross_val_score(knn_cv, x, y, cv=5)

#print the scores of each fold and the mean accuracy
print(cv_scores)
print('cv_scores mean:{}'.format(np.mean(cv_scores)))

```

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection_split.py:670: UserWarning: The least populated class in y has only 1 members, which is less than n_splits=5.

warnings.warn("The least populated class in y has only %d"

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

estimator.fit(X_train, y_train, **fit_params)

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

estimator.fit(X_train, y_train, **fit_params)

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

estimator.fit(X_train, y_train, **fit_params)

In [20]:

```

from sklearn.model_selection import GridSearchCV

knn2 = KNeighborsClassifier()

#create a range for all the values of k we want to test for
param_grid = {'n_neighbors':np.arange(1,30)}

#now use this grid search to test all values
knn_gscv = GridSearchCV(knn2, param_grid, cv=25)

#make the data fit
knn_gscv.fit(x, y)

```

```

C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection\_split.py:670: UserWarning: The least populated class in y has only 1 members, which is less than n_splits=25.
  warnings.warn("The least populated class in y has only %d"
C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection\_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  estimator.fit(X_train, y_train, **fit_params)
C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection\_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  estimator.fit(X_train, y_train, **fit_params)
C:\Users\kieran\anaconda3\lib\site-packages\sklearn\model_selection\_validation.py:531: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  estimator.fit(X_train, y_train, **fit_params)

```

In [21]:

```

#the best n_neighbor for k
knn_gscv.best_params_

```

Out[21]:

```
{'n_neighbors': 17}
```

In [22]:

```

#best possible accuracy for knn
knn_gscv.best_score_

```

Out[22]:

```
0.8935276393230268
```

In [23]:

```

from nltk.stem import WordNetLemmatizer

```

In [24]:

```
#import data on inactive users now
dataIA = 'D:/AI Course/AI Capstone Project/Inactive Accounts.csv'
```

In [25]:

```
DataIA = pd.read_csv(dataIA)
```

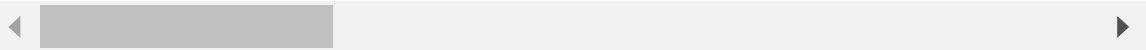
In [26]:

```
DataIA
```

Out[26]:

	Section	Department	AccountExpires	LastLogon	Position	WorkforceCategory
0	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant	PC
1	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant	PC
2	Children's Social Care	Children's Services	NaN	05/05/2023 10:08	Social Worker	NaN
3	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher	LV
4	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher	LV
...	...	...	...	...	...	...
570	NaN	NaN	NaN	NaN	NaN	NaN
571	Assessment and Care Management	Adult Services	28/04/2023 23:00	27/04/2023 13:13	SW Student	LV
572	NaN	NaN	NaN	NaN	NaN	NaN
573	NaN	Public Health	NaN	28/10/2022 13:52	Public Health Administrator	PP
574	Safeguarding (West)	Children's Services	28/07/2023 23:00	17/04/2023 20:59	Locum Social Worker SGCP West 3	CA

575 rows × 40 columns



In [27]:

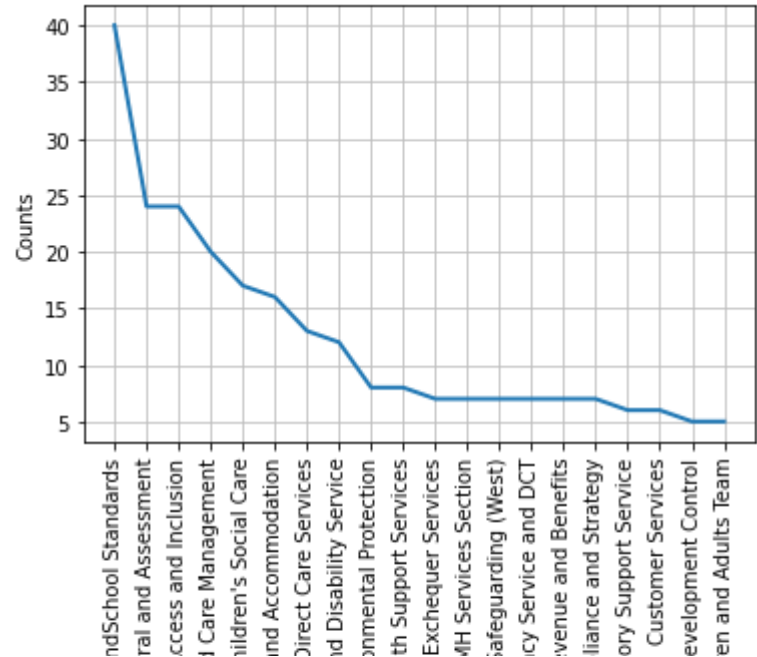
```
#we only want to look at non-NAN accounts as this data set contains temporary staff who
df = DataIA[DataIA['Section'].notna()]
df
```

Out[27]:

	Section	Department	AccountExpires	LastLogon	Position	WorkforceCategory	ManagerPostF
0	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant	PC	1
1	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant	PC	1
2	Children's Social Care	Children's Services	NaN	05/05/2023 10:08	Social Worker	NaN	1
3	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher	LV	1
4	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher	LV	1

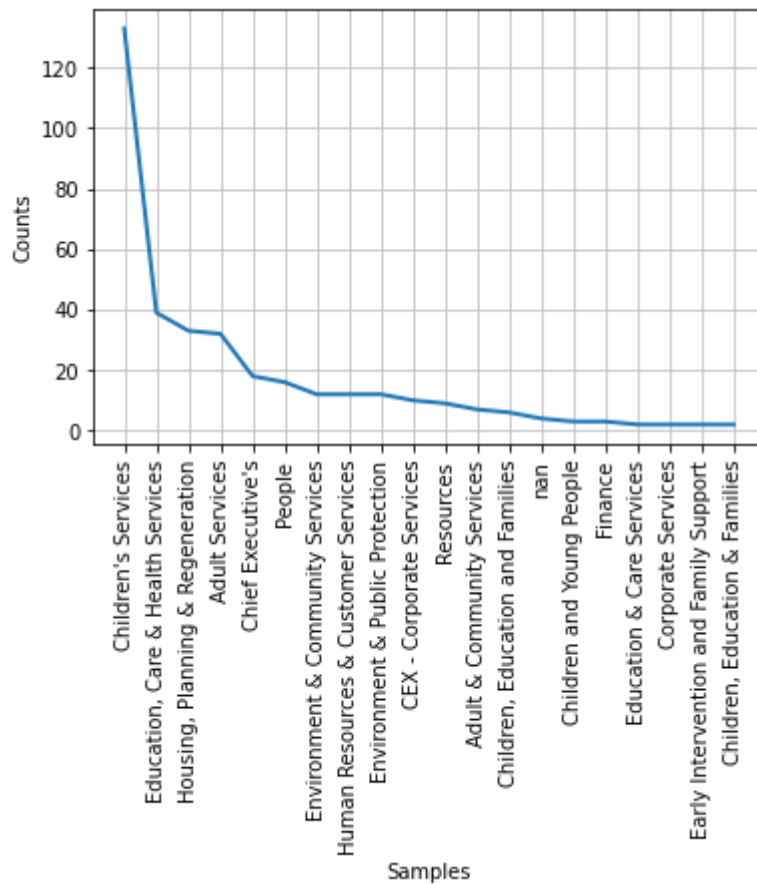
In [28]:

```
#create a frequncy distribution of the Sections
from nltk import FreqDist
fdist = FreqDist(df['Section'])
fdist.plot(20)
```



In [29]:

```
#create a frequency distribution of the Departments  
from nltk import FreqDist  
fdist = FreqDist(df['Department'])  
fdist.plot(20)
```



Out[29]:

```
<AxesSubplot:xlabel='Samples', ylabel='Counts'>
```

In [30]:

```
#Count up users per department  
df.value_counts('Department')
```

Out[30]:

Department	
Children's Services	133
Education, Care & Health Services	39
Housing, Planning & Regeneration	33
Adult Services	32
Chief Executive's	18
People	16
Environment & Public Protection	12
Human Resources & Customer Services	12
Environment & Community Services	12
CEX - Corporate Services	10
Resources	9
Adult & Community Services	7
Children, Education and Families	6
Finance	3
Children and Young People	3
Environmental Services	2
Education & Care Services	2
Early Intervention and Family Support	2
Corporate Services	2
Children, Education & Families	2
People's Department	2
Education, Care and Health Services	1
Facilities	1
Education and care Services	1
Children's Services	1
Place	1
Renewal & Recreation	1
Youth Support Worker in Charge	1

dtype: int64

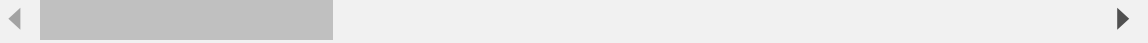
In [31]:

```
df
```

Out[31]:

	Section	Department	AccountExpires	LastLogon	Position	WorkforceCategory	⌵
0	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant		PC
1	Electoral Services	CEX - Corporate Services	28/07/2023 23:00	16/08/2022 11:18	Electoral Assistant		PC
2	Children's Social Care	Children's Services	NaN	05/05/2023 10:08	Social Worker		NaN
3	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher		LV
4	Standards and Attainment	Children and Young People	NaN	14/04/2023 09:22	Inclusion Support Outreach Teacher		LV
...	...	...	...	...	...		...
564	Children's Social Care	Children's Services	NaN	30/01/2023 16:43	Placement Officer		LV
566	Penge and Beckenham Locality Team	People's Department	NaN	17/03/2023 18:23	Care Manager		Wrong
567	Penge and Beckenham Locality Team	People's Department	NaN	17/03/2023 18:23	Care Manager		Wrong
571	Assessment and Care Management	Adult Services	28/04/2023 23:00	27/04/2023 13:13	SW Student		LV
574	Safeguarding (West)	Children's Services	28/07/2023 23:00	17/04/2023 20:59	Locum Social Worker SGCP West 3		CA

368 rows × 40 columns



In [32]:

```
from scipy.stats import chisquare
```

